SK AI SUMMIT 2025 AI NOW & NEXT

A-EYE: Can Large Language Models be an Effective Blind Helper?

Deogyong Kim, Junseong Lee, Junhyeong Park

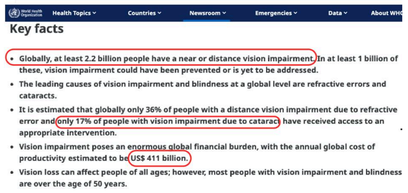
YOUTUBE

DLI LAB

Yonsei Univ. Multimodal AI Lab

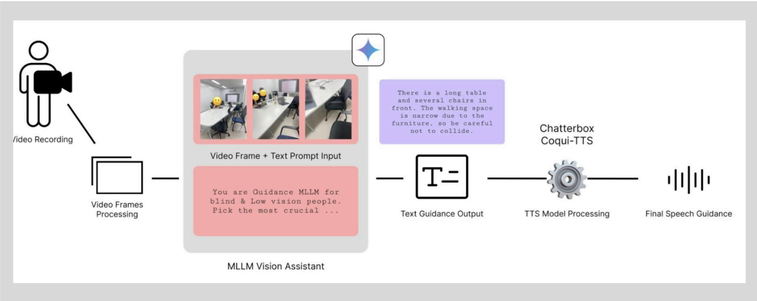
Introduction

cite: Youtube '원샷한솔'



- There are 2.2 billion people worldwide who are visually impaired. Of those, only **17% have access to treatment.**
- Appropriate installation rate of tactile blocks: **4%**, bollards: **4%**, and tactile blocks connected to traffic facilities: **7.8% (Korea)**
- Barriers for the blind and low vision are not personal - they are societal problems **we must solve together.**

A-EYE Pipeline

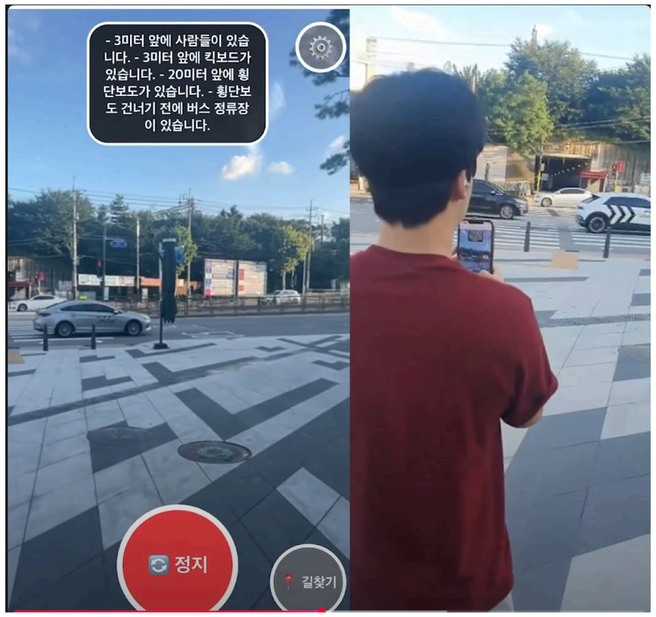


Model	Image Understanding Performance	Throughput (tokens/sec)	Latency (s)
Gemini 1.5 Flash	MMMU 56.1% MathVista 58.4%	163.6	0.21
Gemini 1.5 8B	(Not disclosed)	269.5	0.21
Gemini 2.0 Flash	**MMMU 70.7%** **Vibe-Eval 55.4%+**	**169.5**	**0.53**
Gemini 2.5 Pro	MMMU 72.9% Vibe-Eval 51.3% MMLU ≈ 86.2%	145.2	36.47

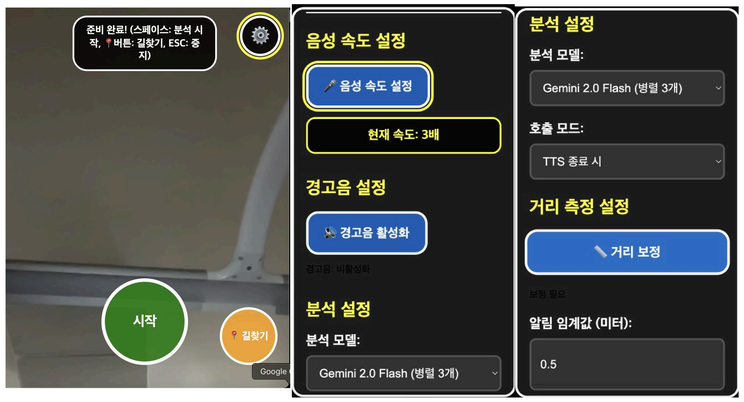
Model (Gemini)	Total latency	API call	File receive	PIL conversion
1.5 Flash	5.62s	4.76s	0.2ms	0.4ms
1.5 8b	4.89s	4.36s	0.4ms	2.4ms
2.0 Flash	**5.32s**	**4.90s**	**0.2ms**	**0.4ms**
2.0 Flash Lite	5.45s	4.25s	0.4ms	0.4ms
2.5 Pro	19.85s	19.31s	0.2ms	2.5ms
2.5 Flash Lite	4.62s	3.08s	0.3ms	0.4ms

- Gemini 2.0 FLASH was the only model with **free usage** and **low latency**. The image understanding performance was also **comparable with pro models.**

Live Demo



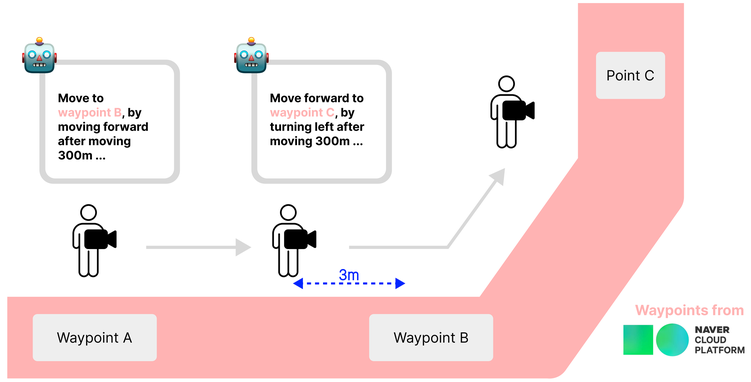
- A-EYE is an AI-Based "**Real-Time Environmental Awareness System**" for the Visually Impaired.
- The system was tested within the campus to determine its effectiveness and was successfully tested.
- Through **human evaluation**, we received feedback that A-EYE offers **precise, real-time scene understanding** and **reassuring safety alerts** during emergencies.



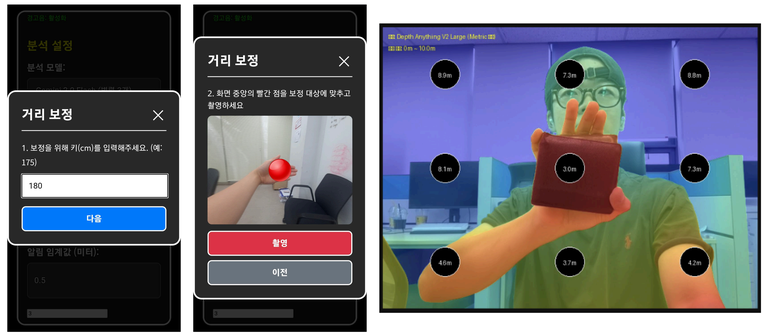
- The A-EYE system can also adjust the TTS voice speed with **voice commands** through setting adjustments.
- This is a measure to **ensure the usability of the service** so that it can respond to the situations of various visually impaired people as much as possible.

Ablations

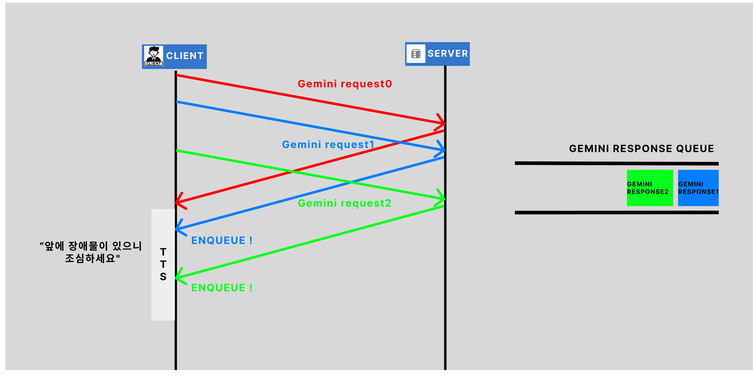
1. Navigation Pipeline (Naver Cloud API)



2. Emergency Alert (Depth Anything v2)



3. Multi-MLLM pipelining



[Citations]
Junhyeok Kim, Jaewoo Park, Junhee Park, Sangeyl Lee, Jiwan Chung, Jisung Kim, Ji Hoon Joung, & Youngjae Yu. (2025). GuideDog: A Real-World Egocentric Multimodal Dataset for Blind and Low-Vision Accessibility-Aware Guidance.
https://www.who.int/docs/default-source/documents/publications/world-vision-report-accessible.pdf?sfvrsn=223f9bf7_2
https://www.ablenews.co.kr/news/articleView.html?idxno=212498
Gemini 2.5 Technical Report
https://developer.mozilla.org/en-US/